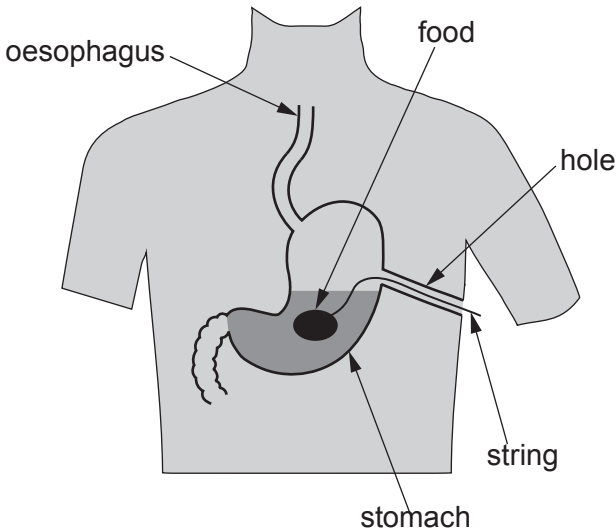


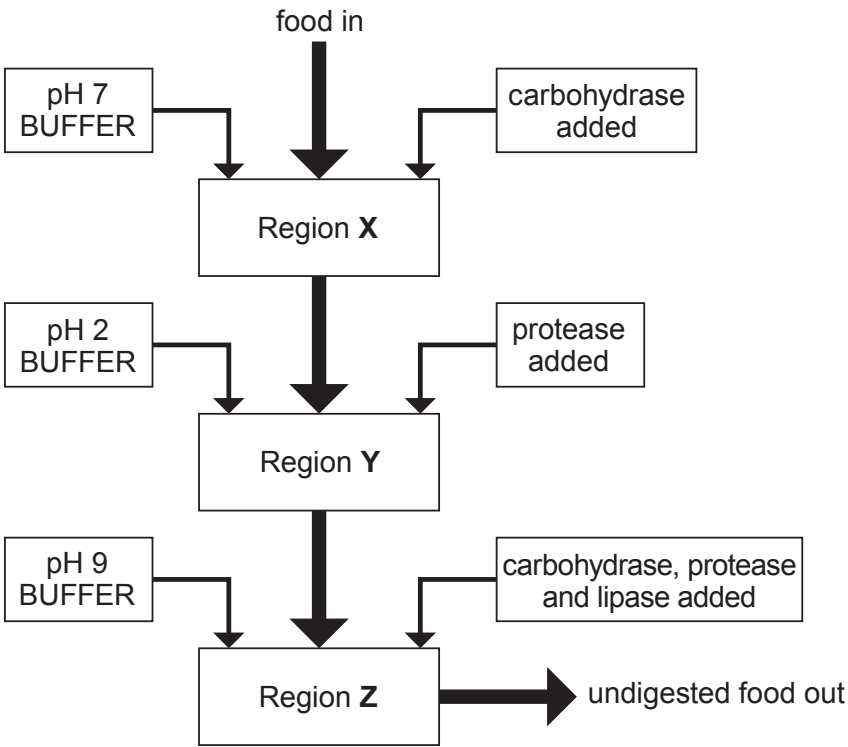
6. On June 6, 1822 Alexis St. Martin, was accidentally shot in the stomach from close range. Dr. William Beaumont treated his wound, but expected Alexis to die from his injuries. Alexis survived, but with a hole through his abdomen wall into his stomach that never fully healed.
- Dr. Beaumont began to carry out experiments on digestion by tying a piece of food to a string and inserting it through the hole into Alexis' stomach. Every few hours, Beaumont would remove the food and assess how much digestion had happened. Beaumont also extracted a sample of gastric juice from the stomach. Analysis showed that the gastric juice was acidic.

Fig 1 – Diagram showing Dr. Beaumont's experiment



Scientists can now follow the digestion of food in detail by using an artificial gut. The diagram below shows how an artificial gut works. (Note: a pH buffer is a chemical that keeps pH constant.)

Fig 2 – Flow chart of an artificial gut



Examiner
only

(a) (i) State the region of the artificial gut that represents the stomach. [1]

.....

(ii) Explain why the pH of each region needs to be different. [2]

.....

(iii) State **one other** factor that would need to be controlled to ensure valid results from an artificial gut. [1]

.....

(iv) State the role of lipase in region **Z**. [1]

.....

(b) One of Dr. Beaumont’s experiments involved inserting meat on a piece of string through the hole in Alexis’ stomach and observing the time taken for the meat to disappear.

Explain why the meat disappeared. [2]

.....

.....

(c) Suggest **two** reasons why scientists prefer to use an artificial gut rather than using human trials. [2]

.....

.....

9

